№8.72

Дано:

$\varepsilon_{вр} = \frac{\hbar^2 l(l+1)}{2I}$

$l \to 2l+1$

$T = 276$ К

NO, $l_m = 7$

Найти: d

Решение:

$$\varepsilon_{вр} = \frac{\hbar^2 l(l+1)}{2I}$$

$$N_l = (2l+1) A e^{-\frac{\hbar^2 l(l+1)}{2IkT}}$$

$$\frac{dN_i}{dl} = 2\cancel{A}\cancel{e^{(\,)}} + (2l+1)\cancel{A}\cancel{e^{(\,)}} \cdot \left(\frac{-\hbar^2(2l+1)}{2IkT}\right) = 0$$

$$2 - \frac{(2l+1)^2\hbar^2}{2IkT} = 0$$

$$4IkT = \hbar^2(2l+l)^2 ; \quad l_m = 7$$

$$4IkT = \hbar^2 \, 15^2$$

$$I = \frac{\hbar^2 15^2}{4kT} = mR^2 = \frac{MR^2}{N_A} \Rightarrow R \approx 1{,}15 \cdot 10^{-10} \text{ м}$$

Ответ: $d = R = 1{,}15 \cdot 10^{-10}$ м.

№8.56

Дано:

$\varepsilon_1 = 0$, $\varepsilon_2 = \varepsilon$

$kT = \varepsilon$

Найти: $C_п$

Решение:

1) $N_i = A e^{-\frac{\varepsilon_i}{kT}}$; $N_1 = A$; $N_2 = A e^{-\frac{\varepsilon}{kT}}$

2) $$\bar{\varepsilon} = \frac{\sum N_i \varepsilon_i}{\sum N_i} = \frac{\cancel{A} e^{-\frac{\varepsilon}{kT}} \cdot \varepsilon}{\cancel{A} + \cancel{A} e^{-\frac{\varepsilon}{kT}}} = \frac{\varepsilon}{e^{\frac{\varepsilon}{kT}} + 1}$$

3) $c = \dfrac{d\mathcal{E}}{dT} = \dfrac{\varepsilon}{\left(e^{\frac{\varepsilon}{kT}}+1\right)^2} \cdot e^{\frac{\varepsilon}{kT}} \cdot \dfrac{\varepsilon}{kT^2} = \dfrac{\varepsilon^2 e^{\frac{\varepsilon}{kT}}}{kT^2\left(e^{\frac{\varepsilon}{kT}}+1\right)^2}$

4) $C_п = \dfrac{3}{2}R + \dfrac{N_A\, \varepsilon^2 e^{\frac{\varepsilon}{kT}}}{kT^2\left(e^{\frac{\varepsilon}{kT}}+1\right)^2} =$

$= \dfrac{3}{2}R + \dfrac{N_A\, k^2T^2 e}{kT^2(e+1)^2} = \dfrac{3}{2}R + \dfrac{Re}{(e+1)^2} = 1{,}7R$

Ответ: $C_п = 1{,}7R$

T9

Дано:
^6Li, T-низкие
$E_п = 0$
$S = 1$; $2S+1$
Найти: S

Решение:

$S = N_A k \ln G$; $G = 3$ $(2S+1)$

$S = R \ln G \approx 9{,}12 \left(\dfrac{Дж}{К\cdot моль}\right)$

№9.8

Дано:
$p = const$
$\nu = 10^{-3}$ моль
Найти: $\dfrac{\Delta T}{T}$

Решение:

$\dfrac{\Delta T}{T}$ – чувствительность опреде-ляется чере флуктуации частиц: $\dfrac{\Delta T}{T} = \dfrac{1}{\;} = \dfrac{1}{\;} \approx 4\cdot10^{-11}$

№10.83

Дано:

$h = 10^{-2}$ м

$\omega = 50$ рад/с

$R = 0{,}1$ м

$f = 100\ \dfrac{\text{дин}\cdot\text{см}}{\text{рад}}$

$\eta = 1{,}8\cdot 10^{-4}$ дин $\dfrac{\text{с}}{\text{см}^2}$

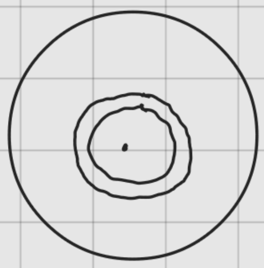
$P = 10^{-4}$ Торр

$v = 450$ м/с

Найти: φ

Решение:

1) Решая задачу 10.82:



$$dF_т = dS \cdot \eta \frac{dv_x}{dy}$$

$$\frac{dv_x}{dy} = \frac{\omega r}{h}$$

$$dF_т = 2\pi r dr \cdot \eta \frac{\omega r}{h}$$

$$dM = 2\pi r dr \eta \frac{\omega r}{h} \cdot r = 2\pi r^3 \eta \frac{\omega}{h} dr$$

$$M = \frac{\pi R^4}{2} \frac{\eta \omega}{h} = \varphi \cdot f \rightarrow \text{в случае } 10.82 \Rightarrow 81^\circ$$

2) При условии $P = 10^{-4}$ Торр: $\eta = \frac{1}{3} \lambda v \rho = \frac{1}{3} h v \rho$

$$\rho = \frac{PM}{RT}; \quad v = \sqrt{\frac{8RT}{M\pi}}; \quad T = \frac{M\pi v^2}{8R}$$

$$\eta = \frac{1}{3} h v \frac{PM}{RT}; \quad v = \sqrt{\frac{8RT}{\pi M}} \Rightarrow T = \frac{\pi M v^2}{8R}$$

$$\eta = \frac{1}{3} h \cancel{v} \frac{P\cancel{M}}{\cancel{R}} \cdot \frac{8\cancel{R}}{\pi \cancel{M} v^{\cancel{2}}} = \frac{8Ph}{3\pi v}$$

$$M = \varphi \cdot f = \frac{\pi R^4}{2} \cdot \frac{\omega}{\cancel{K}} \cdot \frac{8P\cancel{K}}{3\pi \upsilon} \Rightarrow \varphi \approx 1^\circ$$

N10.120

Дано:

$V = 0{,}1\ м^3$

$r = 0{,}02\ м$

$L = 1\ м$

$P_1 = 1\ атм$

$P_2 = 10^{-1}\ мм\,рт\,ст$

$\eta = 1{,}8 \cdot 10^{-5}$

Найти: t

Решение:

1.3.3
ЛАБА

1) $$q = \frac{dm}{dt} = \pi r^2 \rho \upsilon \,;\quad \upsilon = \frac{\upsilon_{max}}{2} = \frac{r^2 \Delta P}{16 \eta L}$$

$$q = \pi r^2 \rho \cdot \frac{r^2}{16\eta} \cdot \frac{P}{L} = \frac{\pi r^4}{16 \eta L} \frac{P^2 M}{RT}$$

2) $$q = -\frac{dm}{dt} = -\frac{\frac{dP}{dt} V \cancel{M}}{\cancel{RT}} = \frac{\pi r^4 P^2 \cancel{M}}{16 \eta L \cancel{RT}}$$

$$\int_{P_1}^{P_2} -\frac{dP}{P^2} = \int_0^t \frac{\pi R^4\, dt}{16 \eta L V}$$

$$\frac{1}{P_2} - \frac{1}{P_1} = \frac{\pi R^4 t}{16 \eta L V} \Rightarrow t = \frac{16 \eta L V}{\pi R^4} \left(\frac{1}{P_2} - \frac{1}{P_1} \right) = \underline{4{,}3\ с}$$

N10.120.

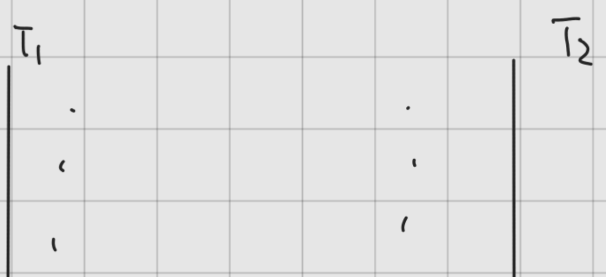
Дано:

T_1, T_2, n

m

Найти: q

Решение:



1) $j = \frac{1}{4} n \bar{v}$; $j_1 = \frac{1}{4} n_1 v_1$; $j_2 = \frac{1}{4} n_2 v_2$

$$j_1 = j_2 \Rightarrow \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\sqrt{T_1}}{\sqrt{T_2}}$$

$$n_1 + n_2 = n = n_1\left(1 + \sqrt{\frac{T_1}{T_2}}\right) \Rightarrow n_1 = \frac{n}{\left(1 + \sqrt{\frac{T_1}{T_2}}\right)}$$

2) $Q^{+} = \frac{1}{6} n_1 v_1 St \, \frac{3k}{2} \cdot T_1$

$$q = \frac{1}{6} \frac{3k}{2} \left(n_1 v_1 T_1 - n_2 v_2 T_2 \right) =$$

$$= \frac{1}{6} \frac{3k}{2} \frac{\sqrt{8k}}{\sqrt{\pi m}} \left(\frac{n}{\left(1+\sqrt{\frac{T_1}{T_2}}\right)} \cdot T_1^{\frac{3}{2}} - \frac{n}{1+\sqrt{\frac{T_1}{T_2}}} \cdot \sqrt{\frac{T_1}{\cancel{T_2}}} \, T_2^{\cancel{\frac{3}{2}}} \right)$$

$$= \frac{\frac{1}{2} k \sqrt{\frac{2k}{\pi m}} \, n}{\sqrt{T_2} + \sqrt{T_1}} \left(T_1^{\frac{3}{2}} \sqrt{T_2} - \sqrt{T_1} \cdot T_2^{\frac{3}{2}} \right) =$$

$$= \frac{\frac{1}{2} k \sqrt{\frac{2k}{\pi m}} \, n}{\cancel{\sqrt{T_2} + \sqrt{T_1}}} \sqrt{T_2 T_1} \cdot \left(\overset{\left(\sqrt{T_1} - \sqrt{T_2}\right)}{\cancel{T_1 - T_2}} \right)$$

$$= \frac{1}{2} \sqrt{\frac{2k^3 T_1 T_2}{\pi m}} \, n \left(\sqrt{T_1} - \sqrt{T_2} \right)$$